Application of Statistical Pattern Recognition in a Generating-Element Space for Human Activity Classification Using Smartphone Sensors

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Abstract. Human Activity Recognition (HAR) using smartphone sensors has become a key component in health monitoring, context-aware computing, and smart environments. Traditional machine learning approaches often assume a Gaussian distribution of sensor data, which may not hold true in real-world scenarios. This paper presents a novel application of the mathematical framework of decomposition in a space with a generating element to address problems in statistical pattern recognition. This approach, based on Kunchenko's theory of stochastic polynomials, enables the effective processing of non-Gaussian data by constructing optimal polynomial features based on the minimization of the mean square error of the decomposition. We apply this method to the UCI HAR dataset, which contains accelerometer and gyroscope data from 30 individuals performing six daily activities. Our approach generates polynomial features of various orders that capture higher-order statistical dependencies in the sensor signals. Experimental results demonstrate that augmenting traditional time-domain and frequency-domain features with these polynomial features improves classification accuracy when using a SVM classifier. The method is particularly effective in distinguishing between static activities (such as sitting and standing), which are traditionally challenging to classify. The proposed approach offers a theoretically grounded feature engineering technique that enhances HAR performance while maintaining computational efficiency suitable for real-time applications.

Keywords: statistical pattern recognition, human activity recognition, smartphone sensors, polynomial features, machine learning, space with a generating element.

1 Introduction

Human Activity Recognition (HAR) with smartphone sensors enables continuous and unobtrusive monitoring for health, safety, and context-aware applications [1]. In real-world settings, accelerometer and gyroscope signals frequently deviate from Gaussian

assumptions because of motion variability, sensor noise, and environmental artifacts—conditions that can degrade discrimination, particularly for activities with similar kinematics (e.g., static poses like sitting vs standing).

To address this, we apply statistical pattern recognition via decomposition in a space with a generating element (DSGE)—a framework grounded in Kunchenko's theory of stochastic polynomials [2,3]. Prior studies demonstrated DSGE's potential to reduce feature variances and improve classification in signal processing, probabilistic diagnostics, and hypothesis testing [4–6]. Here we adapt DSGE for HAR: compact polynomial features are constructed by minimizing mean-square reconstruction error relative to a class-specific generating element and then used with a conventional classifier. We evaluate on the UCI HAR dataset to quantify gains and analyze separation of static activities [7].

2 Literature Review

Classical pipelines and deep learning. Early activity recognition used thresholding and hand-crafted time/frequency features; a classical milestone is Bao & Intille's user-annotated accelerometry study [8]. Broad surveys/tutorials systematized feature-engineering choices and sensor setups for body-worn inertial data [9]. Over the last decade, deep learning became the dominant paradigm: comprehensive reviews cover CNN/LSTM architectures and multimodal designs [10,19]. Canonical models include DeepConvLSTM combining convolutional and recurrent layers [11], hybrid CNN–LSTM pipelines with strong benchmark results [12], and smartphone-focused deep models outperforming classical methods [13]. Comparative studies highlight the role of feature selection and temporal modeling [14–18]. At the same time, many pipelines implicitly assume (near-)Gaussian sensor distributions—assumptions that often fail in free-living data and can reduce separability for look-alike activities.

Recent trends (2020–2025). Two strands are especially relevant to modern HAR:

(i) Self-supervised learning (SSL): pretraining on large unlabeled motion corpora (e.g., UK Biobank-scale accelerometry) improves generalization across devices and free-living conditions and lifts downstream performance versus supervised baselines [20]. A 2024 survey organizes SSL objectives and transfer for accelerometer-based HAR, consolidating techniques for contrastive/predictive pretext tasks and domain adaptation [21].

(ii) Transformer-based encoders: attention mechanisms model long-range temporal dependencies in multi-sensor streams; recent conference work demonstrates transformer-style HAR models and discusses their deployment trade-offs in smart-home/mobile contexts [22].

Related mathematical framework. Independent of deep pipelines, DSGE (Kunchenko space) provides an interpretable, lightweight feature-engineering mechanism tailored to non-Gaussian signals [2–6]. Our approach leverages these properties and complements both classical learners and modern deep encoders.

3 Research Methodology

3.1 Brief theoretical principles

The method of statistical pattern recognition in space with a generating element (Kunchenko space) is based on the concept of decomposing random variables (whose sample is interpreted as a multidimensional vector) using a system of polynomial functions. Contrary to classical methods, which assume a Gaussian distribution of data, this mathematical apparatus allows one to work effectively with any distribution [2,3].

Let X be a vector with arbitrary distribution of components x. The main idea of the method is to approximate X as follows:

$$X = k_0 + \sum_{i=1}^{S} k_i \, \varphi_i(X) + Z, \tag{1}$$

where

- $k_0, k_1, \ldots, k_s$ - optimal decomposition coefficients;
- $\varphi_i(\xi)$ - a set of nonlinear functions;
- Z - approximation error;
- S - order of approximation.

The key feature of the method is minimising the mean square error of the decomposition (MSED):

$$D = E\{Z^2\} \rightarrow min. \tag{2}$$

The optimal coefficients k_i are found from a system of linear algebraic equations:

$$\boldsymbol{K}\,\boldsymbol{F} = \boldsymbol{B}, \tag{3}$$

where $\boldsymbol{F}$ is a square matrix of dimension S consisting of centered correlants $F_{i,j} = E\{[\varphi_i(x) - \Psi_i][\varphi_j(x) - \Psi_j]\}$, $\boldsymbol{B}$ - vector of mutual correlations $B_i = E\{[x - \Psi_0][\varphi_i(x) - \Psi_i]\}$, which are dependent on the mathematical expectations of the basic transforms $\Psi_i = E\{\varphi_i(x)\}$, a $\Psi_0 = E\{x\}$.

An additional condition for minimising criterion (2) is the requirement for coefficient k_0, which must be found as:

$$k_0 = \Psi_0 + \sum_{i=1}^{S} k_i \, \Psi_i. \tag{4}$$

Various functions can be used as bases for the decomposition (1): power, trigonometric, logarithmic, and others.

3.2 Justification and usage methodology

The analysis of DSGE properties provides a universal way to transform a variety of functional objects – from statistical data, time series or space slices to spectral profiles – into a compact and informative vector of parameters suitable for further classification processing. The core idea behind this conversion is to use a linear combination of nonlinear basis functions $\varphi_i(X)$ to form differences between the generating element and its approximation. By virtue of its coherence properties, this difference represents the

appropriate class, characterized by a certain set of parameters Ψ_i. That is why the basis ceases to be 'universal' in the sense of orthogonal transformations (e.g., Fourier or wavelet analysis) and acquires a statistical 'link' to the structural features of a particular class.

The methodology of using DSGE for constructing classifiers can be represented as a sequence of the following steps:

1. The procedure for forming DSGE features begins with the selection of an adequate functional representation of the source data. For time signals, it is natural to represent them as vectors obtained by discretisation *x(t)*; for images – one-dimensional profiles or two-dimensional slices; for spectral data – amplitude-frequency dependencies; and for text – vector representations of words or individual tokens.
2. Next, it is necessary to define a compact set of independent functions that form polynomials of relatively low orders (1), the type of which is selected based on considerations of effective minimisation of MSED (2) and computational simplicity. The resulting generating element can be defined as the average or reference signal of a certain class, thus forming a geometric 'anchoring' of the subspace in which the projection will be performed.
3. The next step is to solve the system of normal equations (3), which may require a small Tikhonov regularisation, the use of which is necessary to ensure the stability of estimates in conditions of a limited training sample and the presence of noise. The resulting vector of optimal coefficients **K** will reflect the consistency of the object under study with the selected basis and parent element, and the quadratic error of object reconstruction through projection will serve as an integral characteristic of its belonging to the corresponding class.

From the point of view of statistical pattern recognition theory, the advantage of DSGE lies in the fact that it combines the properties of principal component analysis and metric learning. On the one hand, linear projection reduces dimensionality and filters out high-frequency noise, and on the other hand, the generating element creates a natural cluster structure: a set of reconstruction error vectors relative to different classes immediately forms a distance space in which traditional algorithms such as SVM, logistic regression or k-NN demonstrate increased discriminative power even on small samples. It is important to emphasise that all further calculations boil down to scalar product operations and inversion of a small S×S matrix, i.e. they can be implemented on embedded processors with strict resource limitations without significantly degrading response time.

It should be noted that the presented methodology for using DSGE is universal and can be applied to solve a wide range of tasks: from the analysis of biomedical signals, classification of sensor data, to technical diagnostics of the condition of aggregates based on vibration data and recognition of intents in text messages. In all these cases, objects of different classes have a distinct shape or dynamics that can be successfully modelled by the selected basis, while the generating element captures the key 'average' morphology of the class It is this combination of adaptability and interpretability that potentially allows this approach to act as a single coherent paradigm for the formation of classification features that can be effectively integrated into both classical algorithms

(SVM, Random Forest) and special layers of neural networks — provided that sufficient data and computing resources are available. In the next section, it will be demonstrated how the described methodology is implemented and evaluated on specific experimental tasks, the results of which confirm the practical workability of the proposed approach.

4 Results and Discussion

4.1 Description of experimental data

To validate the proposed method, a series of experiments was conducted on the public UCI Human Activity Recognition Using Smartphones dataset [7]. This dataset contains records from 30 volunteers aged 19 to 48 who performed six basic activities of daily living: walking (WALKING), going up stairs (WALKING_UPSTAIRS), going down stairs (WALKING_DOWNSTAIRS), sitting (SITTING), standing (STANDING) and lying down (LAYING).

The data was collected using a Samsung Galaxy S II smartphone attached to a belt, with a sampling rate of 50 Hz. Each recording contains 9 signal channels:

- 3-axis body accelerometer (body_acc_x, y, z);
- 3-axis gyroscope (body_gyro_x, y, z);
- 3-axis total acceleration (total_acc_x, y, z).

The signals were segmented into 2.56-second windows (128 samples) with 50% overlap. The dataset was pre-divided into training (7352 instances, 70%) and test (2947 instances, 30%) samples. All experiments were conducted using the Python 3.13 programming language and the Scikit-learn framework. The full code for the experiments is available in the project repository at: [https://github.com/SZabolotnii/SKPR_HAR].

4.2 Methodology of experiments

Data pre-processing and formation of a generating signal

Before applying DSGE, all signals were scaled for each individual channel across the entire training sample in order to normalise the data (reducing the mathematical expectation to 0 and the variance to 1). Thus, the generating element in the sense of (1)–(4) for each class was the *average normalised signal* of the corresponding activity type.

Formation of basis functions and calculation of decomposition coefficients

In accordance with the theoretical foundations of DSGE (Section 3), separate reconstruction models were trained for each activity class (with the formation of corresponding sets of parameters Ψ_i). Four different types of bases were used (the results are presented in Table 1). In the process of searching for optimal coefficient values, Tikhonov regularisation was used with a regularisation parameter $\lambda = 0.01$, the value of which was chosen empirically to ensure the stability of the solution of the system of equations of type (3).

Classification feature generation

Based on the trained models, a vector of 6 features (for each class) was calculated for each test sample - logarithms of aggregated mean square reconstruction errors:

$$\varepsilon_c = \log \sum_j \left\| \tilde{x}_j - \hat{x}_j^c \right\|^2, \tag{5}$$

where $c \in \{0,...,5\}$ - class index, j - channel index.

4.3 Experiment results

Experiment 1: Basic application of DSGE

In the first experiment, the efficiency of the proposed method was evaluated using only 6 generated features. The support vector machine (SVM) method with an RBF kernel was used for classification. In the basic configuration, an unoptimised version was used with a modified power ($S=4$) functional basis: $sign(x)|x|^2$, $sign(x)|x|^3$, $sign(x)|x|^4$.

Results:

- Overall accuracy: **77.71%**.
- Excellent recognition of the LAYING class (100% accuracy).
- Good differentiation between static poses SITTING (79% F1-score) and STANDING (82% F1-score).
- The hardest thing was telling apart dynamic activities, especially WALKING_UPSTAIRS (58% F1-score).
- WALKING has the lowest accuracy among all classes (68% F1 score).

Experiment 2: Analysis of different basis functions

To check out how the choice of basis affects things, four types of functions were tested:

Table 1. Types of basis functions and results of their application.

Types of basis	Functions	Accuracy
Polynomial	x^2, x^3, x^4	76.04%
Trigonometric	sin(x), sin(2x), sin(3x)	78.76%
Robust	tanh(x), sigmoid(x), atan(x)	81.40%
Fractional	$\sqrt{\lvert x\rvert}$, $\sqrt[3]{\lvert x\rvert}$, $\sqrt[4]{\lvert x\rvert}$	82.59%
Ensemble (all 4 types)	24 features	**84.63%**

The fractional basis demonstrated the best individual performance among all types. It is important to note that the functions of this $sign(x)|x|^p$ show better results than simple power functions because they retain information about the sign of the signal.

The ensemble of all 4 experts achieved the highest accuracy of 84.63%, correcting a significant number of errors made by individual experts:

- Polynomial: 392 errors corrected;
- Trigonometric: 321 errors corrected;

- Robust: 242 errors corrected;
- Fractional: 228 corrected errors.

Experiment 3: Optimisation of basis parameters

To determine the optimal parameters of the optimized fractional basis (Optimized Fractional) $sign(x)|x|^p$, a grid search was performed for parameters S: [2, 3, 4, 5] (number of basis functions). The training sample was divided into training (5881 samples) and validation (1471 samples) parts.

Optimisation results:

- Best parameter value: S=3.
- Validation accuracy: 83.41%.
- Test accuracy: 84.36%.

It is important to note that the obtained optimal variant allowed us to achieve a result that is practically equal to the ensemble (see Table 1), but using only 3 basis functions instead of 12.

Experiment 4: Hybrid model

To improve the results, a hybrid models was developed that combines 6 polynomial features with 24 of the most informative traditional features (selected using Random Forest).

Hybrid model results:

- Overall accuracy: **88.46%**
- Cross-validation accuracy: **89.58%**
- Optimal SVM parameters SVM: C=100, γ=0.01

A detailed analysis by class is presented in Table 2:

Table 2. Classification accuracy metrics for the hybrid model.

Class	Accuracy	Recall	F1-score
WALKING	0.88	0.96	0.92
WALKING_UPSTAIRS	0.84	0.87	0.85
WALKING_DOWNSTAIRS	0.95	0.83	0.89
SITTING	0.86	0.75	0.80
STANDING	0.80	0.89	0.84
LAYING	1.00	1.00	1.00

Analysis of the results obtained shows the dominance of gravitational components and time domain features, confirming their importance for human activity recognition. The highest accuracy is achieved for the LAYING class (100%), the lowest for the static poses SITTING and STANDING.

Experiment 5: Incremental validation

To prove the unique informational value of the proposed method, an experiment was conducted with the addition of 6 MSED features to the complete set of 561 traditional features. The purpose of this experiment was to determine whether the new features

contain unique information not represented in the standard UCI HAR set.

Results of the basic model (561 features):

- Accuracy: **95.72%.**
- Very high output performance due to a comprehensive set of traditional features.

Results of the extended model (567 features = 561 traditional + 6 polynomial):

- Accuracy: **95.86%**

Analysis of corrected errors: Polynomial features corrected 7 classification errors, all related to the distinction between static poses:

- SITTING ← STANDING: 5 *corrections.*
- STANDING ← SITTING: 2 *corrections.*

Interpretation of results: Although the improvement is small, the experiment demonstrates several important aspects:

1 **The 'ceiling' effect on performance is evident**: With a baseline accuracy of 95.72%, further improvement becomes extremely challenging due to the approach of the theoretical maximum.
2 **Specificity of improvements**: All corrected errors relate to the differentiation between static poses (SITTING/STANDING), confirming the ability of the new features to capture subtle differences in these difficult-to-recognise activities.
3 **Complementarity of information**: Even with 561 high-quality traditional features, polynomial features still reveal additional patterns, albeit to a limited extent.
4 **Methodological limitations**: The dataset used already contains a highly developed set of engineering features, which reduces the scope for further improvements.

4.4 Analysis of computational complexity

The proposed method has the following complexity characteristics:

Training phase:

- Calculation of basis functions: $O(N \cdot T \cdot S)$.
- Solving systems of equations: $O(M \cdot C \cdot S^3)$.
- Overall complexity: $O(N \cdot T \cdot S + M \cdot C \cdot S^3)$.

where N is the number of samples, T is the length of the time window, S is the number of basis functions, M is the number of classes, and C is the number of channels.

Classification phase:

- Feature generation: $O(T \cdot S \cdot M \cdot C)$.
- SVM prediction: $O(M \cdot Nsv)$.

where Nsv is the number of support vectors.

For the optimal configuration ($S=3, M=6, C=9$), the method is computationally efficient and can run in real time on mobile devices. The hybrid model with 30 features remains compact enough for practical application. Even the extended set of 567 features retains acceptable computing complexity for offline analysis tasks.

5 Conclusions

This paper is the first to explore the application of statistical pattern recognition in space with a generating element to solve the problem of human activity recognition using smartphone sensors. A new feature generation methodology has been developed, based

on minimising the mean square error of signal reconstruction through optimal fractional-power bases. The optimal basis parameters have been experimentally established, providing an accuracy of 84.36% using only 6 features. A hybrid approach combining polynomial features with traditional methods achieves the highest accuracy of 88.46%, demonstrating the synergistic effect and complementarity of the proposed method.

The theoretical contribution of the study lies in substantiating the effectiveness of fractional power functions for modelling non-Gaussian distributed sensor data and expanding the scope of application of Kunchenko's theory of stochastic polynomials. The practical significance of the research is determined by the creation of a computing-efficient method with linear complexity relative to the signal length, which makes it suitable for real-time applications on mobile devices. The method demonstrates exceptional effectiveness in distinguishing static activities and can be easily integrated with existing activity recognition systems.

The results obtained open up new possibilities for applying statistical pattern recognition theory in the analysis of signals of various origins and create a basis for the development of energy-efficient monitoring systems. Further research should focus on validating the method on different types of data and integrating it with modern deep learning architectures.

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